

7. (2.5 points)

Given the following information:

	Cumulative Reported Loss (\$000) as of (months)				
Accident Year	12	24	36	48	60
2012	1,300	2,350	3,500	3,650	3,750
2013	1,450	3,000	5,050	5,850	
2014	1,600	2,500	3,700		
2015	1,750	3,050			
2016	1,950				

	Observed Loss Development Factors as of (months)			
Accident Year	12-24	24-36	36-48	48-60
2012	1.808	1.489	1.043	1.027
2013	2.069	1.683	1.158	
2014	1.563	1.480		
2015	1.743			

- There is no development beyond 60 months.

a. (0.5 point)

Calculate the ultimate loss for AY 2014 under the assumption that the variance of the next incremental loss is a constant for each development age.

b. (0.5 point)

Calculate the ultimate loss for AY 2014 under the assumption that the variance of the next incremental loss is proportional to the losses reported to date.

c. (0.5 point)

Calculate the ultimate loss for AY 2014 under the assumption that the variance of the next incremental loss is proportional to the square of the losses reported to date.

d. (1 point)

Fully describe a test to determine which of the three variance assumptions in parts a., b., and c. above is most reasonable for the data in this triangle. Do not perform any calculations.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 7										
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE(S): A2									
SAMPLE ANSWERS										
Part a: 0.5 point										
<u>Sample 1</u>										
$36-48 = \frac{1.043 \times 3,500^2 + 1.158 \times 5,050^2}{3,500^2 + 5,050^2} = 1.121$										
48-Ult = 1.027										
2014 Ultimate = $3,700 \times 1.121 \times 1.027 = 4,260$										
<u>Sample 2</u>										
$\text{LDF } 36-48 = \frac{3,500 \times 3,650 + 5,050 \times 5,850}{3,500^2 + 5,050^2} = 1.121$										
$\text{LDF } 48-60 = \frac{3,650 \times 3,750}{3,650^2} = 1.027$										
LDF 36-Ult = $1.121 \times 1.027 = 1.151$										
Ultimate loss = $3,700 \times 1.151 = 4,258.7$										
<u>Sample 3</u>										
Use least squares LDF										
	<table><tr><td></td><td><u>36-48</u></td><td><u>48-60</u></td></tr><tr><td>LDF</td><td>1.121</td><td>1.027</td></tr><tr><td>CDF</td><td>1.151</td><td>1.027</td></tr></table>		<u>36-48</u>	<u>48-60</u>	LDF	1.121	1.027	CDF	1.151	1.027
	<u>36-48</u>	<u>48-60</u>								
LDF	1.121	1.027								
CDF	1.151	1.027								
AY 2014 Ultimate = $3,700K \times 1.151 = 4,260K$										
Part b: 0.5 point										
<u>Sample 1</u>										
$f_{36-48} = \frac{1.043 \times 3,500 + 1.158 \times 5,050}{3,500 + 5,050} = 1.1109$										
AY 2014 Ult = $3,700 \times 1.1109 \times 1.027 = 4,221.4$										
<u>Sample 2</u>										
$\frac{3,650 + 5,850}{3,500 + 5,050} = 1.111$										
$3,700 \times 1.111 \times 1.027 = 4,222$										

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample 3

Use Volume Weighted LDF

	36-48	48-60
LDF	1.111	1.027
CDF	1.141	1.027

$$\text{AY 2014 Ultimate} = 3,700K \times 1.141 = 4,222K$$

Part c: 0.5 point

Sample 1

$$f_{36-48} = \frac{1.043 + 1.158}{2} = 1.101$$

$$3,700 \times 1.101 \times 1.027 = 4,184$$

Sample 2

Use Straight Average LDFs

	36-48	48-60
LDF	1.101	1.027
CDF	1.131	1.027

$$\text{AY 2014 Ultimate} = 3,700K \times 1.131 = 4,182K$$

Part d: 1 point

- To test the variance assumptions, we can plot weighted residuals against $c_{i,k}$

Plot a: $(c_{i,k+1} - c_{i,k} \times f_{k0})$ against $c_{i,k}$

Plot b: $(c_{i,k+1} - c_{i,k} \times f_{k1}) / \sqrt{c_{i,k}}$ against $c_{i,k}$

Plot c: $(c_{i,k+1} - c_{i,k} \times f_{k2}) / c_{i,k}$ against $c_{i,k}$

Compare plots a, b, and c for several values of k to see which shows the most random pattern. That one is the most reasonable for the data in the triangle.

- Calculate the following residuals:

Situation a: $r = (c_{i,k+1} - c_{i,k} \times f_{k0})$

Situation b: $r = (c_{i,k+1} - c_{i,k} \times f_{k1}) / \sqrt{c_{i,k}}$

Situation c: $r = (c_{i,k+1} - c_{i,k} \times f_{k2}) / c_{i,k}$

Plot these r's against previous cumulative losses. If an assumption's r's are random around 0 with no trends or patterns, then that assumption is appropriate. If all three plots are random, choose f_{k1} because the other two are not superior.

EXAMINER'S REPORT

Candidates were expected to know which Chain Ladder weighting method corresponded to each of the Mack variance assumptions. They were also expected to know how to test which variance assumption was most appropriate.

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part a

Candidates were expected to use Mack's Least Squares methodology to generate LDFs. They were also expected to apply the LDFs to generate ultimate losses.

Common mistakes included:

- Using the Brosius Least Squares method.
- Using the simple average variance assumption.
- Using the wrong AY to calculate LDFs.

Part b

Candidates were expected to use the volume weighted average methodology to generate LDFs. They were also expected to apply the LDFs to generate ultimate losses.

Common mistakes included:

- Using an incorrect variance assumption or a different weighting method which wasn't appropriate for the given situation.

Part c

Candidates were expected to use the simple average methodology to generate LDFs. They were also expected to apply the LDFs to generate ultimate losses.

Common mistakes included:

- Using the Least Squares variance assumption, since similar to an OLS regression, Least Squares assumes that variance of the next incremental loss is constant (i.e. variance does not change based on the losses reported to date).
- Using square root or $3/2$ power to weight.

Part d

Candidates were expected to be able to explain Mack's method for testing the proper variance assumption. This appeared to be the challenging part of the question for candidates.

Common mistakes included:

- Describing the linearity test instead.
- Describing a method that did not test the variance assumption (e.g. comparing MSE or adjusted SSE).
- Comparing the values in the triangle for seasonality or outliers.
- Not providing sufficient detail and/or inaccurate detail.